

# STAT 24620 / FINM 34700 / STAT 32950

## Homework 1

Due: 04/05/2026 11:59pm

### Problem 1: Descriptive multivariate analysis

Download the data `ladyrun25.dat` and save the dataset in your working directory. The data are on national track records for women, based on Table 1-9 in Johnson and Wichern. Measurements for 100m, 200m, and 400m are in seconds, longer distance records are in minutes. Variable names are not included.

The following R command can be used to input the data

```
ladyrun = read.table("ladyrun25.dat")
colnames(ladyrun)=c("Country", "100m", "200m", "400m", "800m", "1500m", "3000m", "Marathon")
```

Exclude the nominal variable identifying the country when carrying out numerical calculations.

- Compute the sample means of the numerical variables. Is there any variable for which the sample mean is less interpretable than the others? Briefly explain.
- Compute the sample covariance matrix and the sample correlation matrix.
- Compute the sample correlation matrices based on Spearman's rank correlation and Kendall's  $\tau$ .
- Compare the Pearson, Spearman, and Kendall correlation summaries briefly. In this dataset, do they tell a broadly similar story, or do they suggest meaningful differences? Give a short explanation.
- For the Pearson sample correlation matrix  $\mathbf{R}$ , compute its eigenvalues and eigenvectors.
- What is the sum of the eigenvalues of  $\mathbf{R}$ ? How does this relate to the number of numerical variables in the dataset?
- Briefly interpret what a dominant first eigenvalue would suggest about the dependence structure of the variables.

### Problem 2: Covariance and spectral decomposition

- Let  $\mathbf{Y} \in \mathbb{R}^p$  be a random vector with covariance matrix  $\Sigma$ . Show that  $\Sigma$  is positive semi-definite, i.e., all  $p$  eigenvalues of  $\Sigma$  must be non-negative.

(b) Let

$$A = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \rho \in (0, 1).$$

Find the eigenvalues and unit-length eigenvectors of  $A$ .

- (c) Write out the spectral decomposition  $A = V\Lambda V^\top$ , where the columns of  $V$  are orthonormal eigenvectors, and  $\Lambda$  is the diagonal matrix of eigenvalues of  $A$ .
- (d) Use the spectral decomposition to write  $A^{-1}$  explicitly in terms of matrix operations on  $V$  and  $\Lambda$ .

### Problem 3: PCA analysis on running-record

Use the same running-record dataset as in Problem 1.

- (a) Before carrying out both analyses, give an *a priori* expectation about whether covariance-based PCA or correlation-based PCA is likely to be more appropriate for this dataset. Your answer should refer to the scales/units of the variables. You may revisit this prediction after comparing the two analyses later in the problem.
- (b) Carry out PCA on the standardized data (equivalently, using the correlation matrix). Determine the first two principal components and express them as linear combinations of the standardized variables.
- (c) Produce a scree plot and report the cumulative proportion of variance explained by the first two principal components. Based on these diagnostics, how many components would you retain for a low-dimensional summary? Briefly justify your choice.
- (d) Produce a score plot of the observations in the (PC1, PC2) plane and briefly comment on any visible pattern.
- (e) Produce a loading plot of the original variables in the (PC1, PC2) plane and briefly interpret what PC1 and PC2 appear to represent.
- (f) Repeat the PCA using the covariance matrix instead of the correlation matrix. Compare the leading loading vector, the explained variance pattern, and the practical interpretation with those from part (b)–(e).
- (g) Based on your comparison in part (f), comment on how sensitive the PCA conclusions are to preprocessing choices, and which version you would report in practice.

### Problem 4: Yield-curve PCA and reconstruction

Download the data `treasury_yield_changes_bps.csv`, the same dataset used in Lecture 2. It contains daily Treasury yield levels (in percent) from 2014–2026 for the maturities

$$\{1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y\}.$$

- (a) Convert the yield-level data to daily Treasury yield changes and change the unit to basis points (bps), where 1 bps = 0.01%.

- (b) Apply PCA to the centered daily yield changes in bps, using covariance-based PCA. Report the first three loading vectors and the proportion of variance explained by PC1, PC2, and PC3.
- (c) For a given day  $i$ , let  $\mathbf{x}_{c,i} \in \mathbb{R}^p$  denote the centered yield-change vector, and let  $\mathbf{v}_1, \mathbf{v}_2$  be the first two loading vectors with corresponding scores  $z_{i1}, z_{i2}$ . For the remaining parts of this problem, *define* the rank-2 PCA reconstruction by

$$\hat{\mathbf{x}}_{c,i}^{(2)} = z_{i1}\mathbf{v}_1 + z_{i2}\mathbf{v}_2.$$

Here  $\hat{\mathbf{x}}_{c,i}^{(2)}$  is the fitted cross-maturity yield-change vector for day  $i$  obtained by keeping only the first two principal-component directions. Use this definition and answer the following questions:

- (i) Define the residual vector

$$\mathbf{r}_i = \mathbf{x}_{c,i} - \hat{\mathbf{x}}_{c,i}^{(2)}.$$

Show that  $\mathbf{r}_i$  is orthogonal to both  $\mathbf{v}_1$  and  $\mathbf{v}_2$ . You may use the facts that the loading vectors are orthonormal and that  $z_{ij} = \mathbf{v}_j^\top \mathbf{x}_{c,i}$ .

- (ii) Choose one day  $i$  for which the rank-2 approximation is visibly imperfect—for example, a day with relatively large  $|z_{i3}|$  or, equivalently, a comparatively large residual norm  $\|\mathbf{r}_i\|$ . Using that selected day, use your PCA output to:
  - (1) plot the observed centered yield-change vector for day  $i$  against maturity (that is, the cross-maturity yield-change curve across the 8 Treasury maturities) together with its rank-2 PCA reconstruction  $\hat{\mathbf{x}}_{c,i}^{(2)}$ ;
  - (2) briefly explain what aspect of the daily cross-maturity curve movement is not captured when only two PCs are used.

## Problem 5: factor-analysis on running-record

Using the same running-record dataset as in Problem 1, fit a two-factors factor-analysis model in R using `factanal`.

- (a) Report the estimated loading matrix and uniquenesses from your factor-analysis fit.
- (b) Briefly explain which variables appear to load most strongly on the factor(s), and which variables have relatively large uniqueness.
- (c) Compare the factor-analysis output with the PCA results from Problem 3. In particular, comment on one similarity and one difference between the interpretations you would give from FA and from PCA.